LAB5 – Analysis of Network Services (DNS, HTTP, FTP)

Course: The Bits and Bytes of Computer Networking
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1. Introduction

This laboratory analyzes the operation of multiple network services on a single server within a local network using Cisco Packet Tracer.

The server was configured with **DNS, HTTP, and FTP** services. By using **Simulation Mode**, it was possible to observe the packets exchanged between the client and the server, allowing an understanding of how application layer protocols use transport protocols and specific ports to provide network services.

This demonstrates the interaction between different layers of the OSI model in a client-server communication.

2. Network Topology

The network used in this laboratory consists of:

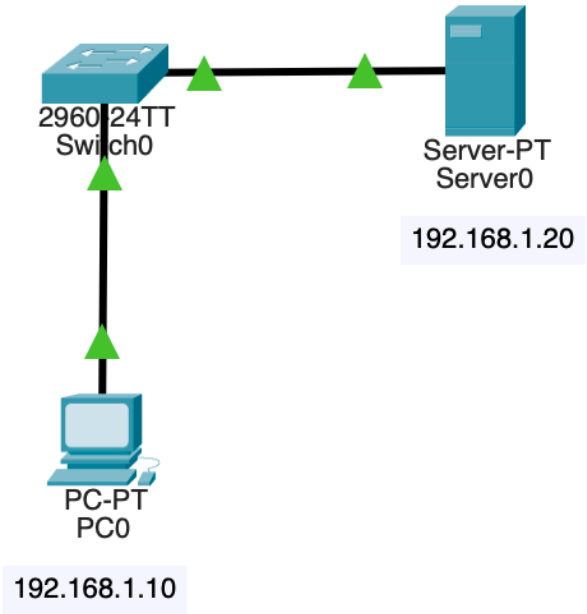
- One PC acting as a client

- One Cisco 2960 switch

- One server providing network services

All devices are connected within the same local area network (LAN) through the switch.

Figure 1 – Network Topology



3. Server Services Configuration

The server was configured to provide multiple network services.
The enabled services were:

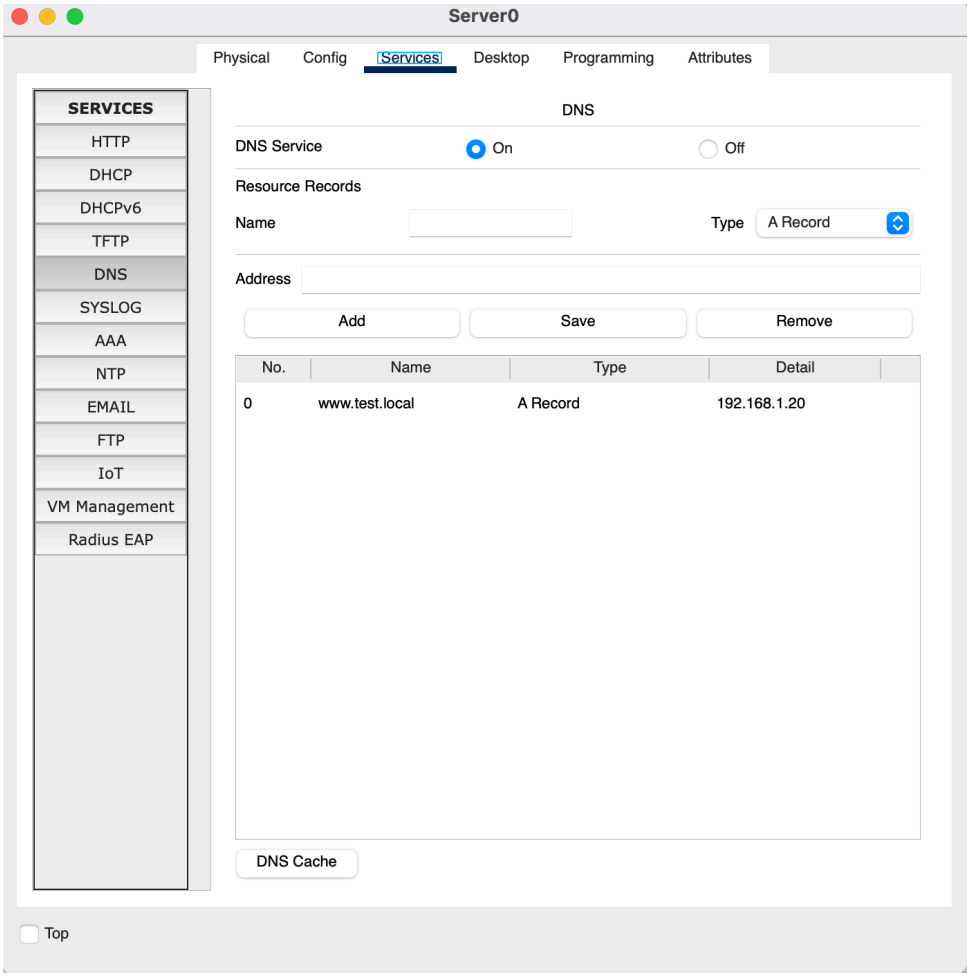
- DNS

- HTTP

- FTP

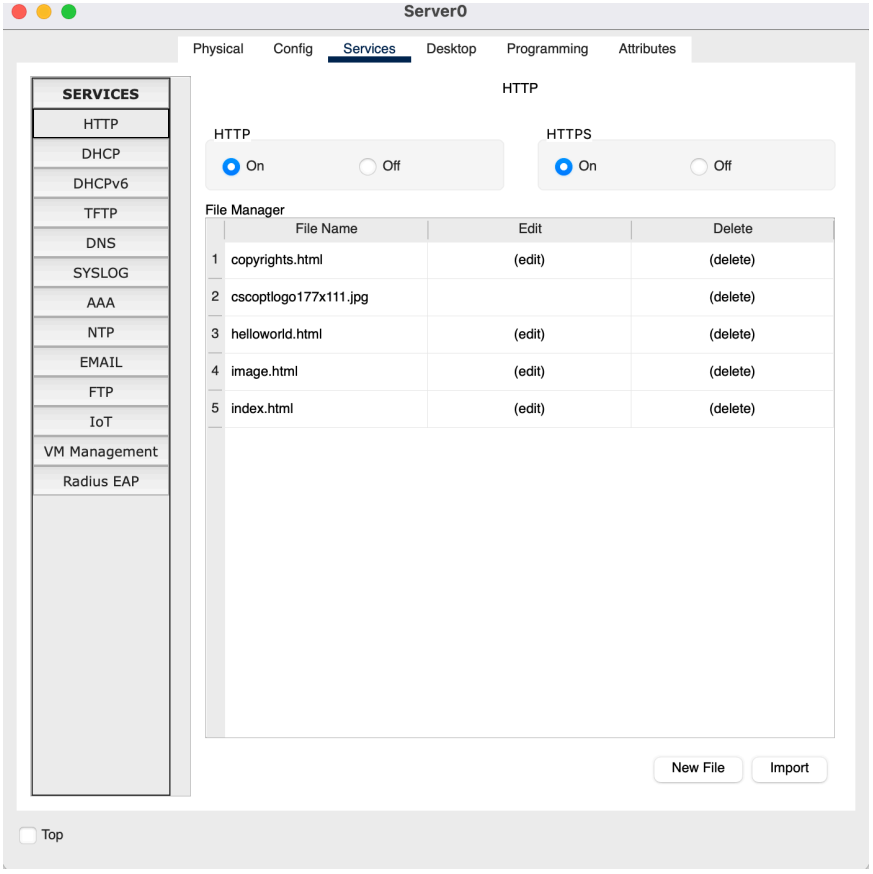
These services allow the client to access the server using a domain name and different application layer protocols, providing name resolution, web content access, and file transfer.

Figure 2 – DNS Configuration



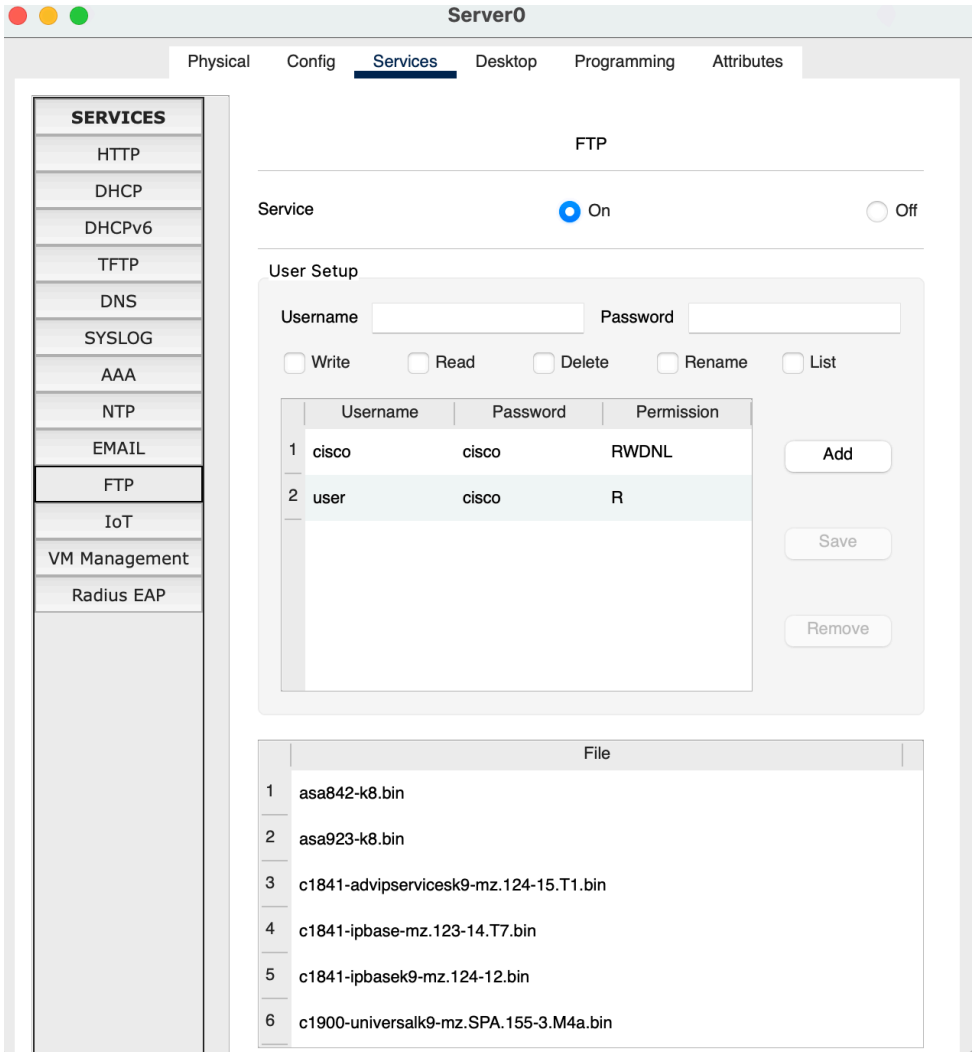
The server was configured to act as a DNS server, allowing the resolution of the domain name **www.test.local** to its corresponding IP address.

Figure 3 – HTTP Configuration



The HTTP service was enabled on the server to allow access to a web page from the client using a web browser.

Figure 4 – FTP Configuration



The FTP service was configured to allow user authentication and file transfer between the client and the server.
This service uses port 21 over TCP for connection control.

4.1 Protocol Analysis

Using Simulation Mode, different protocols involved in the communication between the client and the server were observed.

Protocol	Port	Transport
Domain Name System	53	UDP
Hypertext Transfer Protocol	80	TCP
File Transfer Protocol	21	TCP

4.2 DNS Resolution

Before accessing the web server, the client must resolve the domain name.

The client sends a DNS query requesting the IP address associated with the domain:

www.test.local

The DNS server responds with the corresponding IP address.

The Domain Name System uses:

UDP port 53

4.3 HTTP Communication

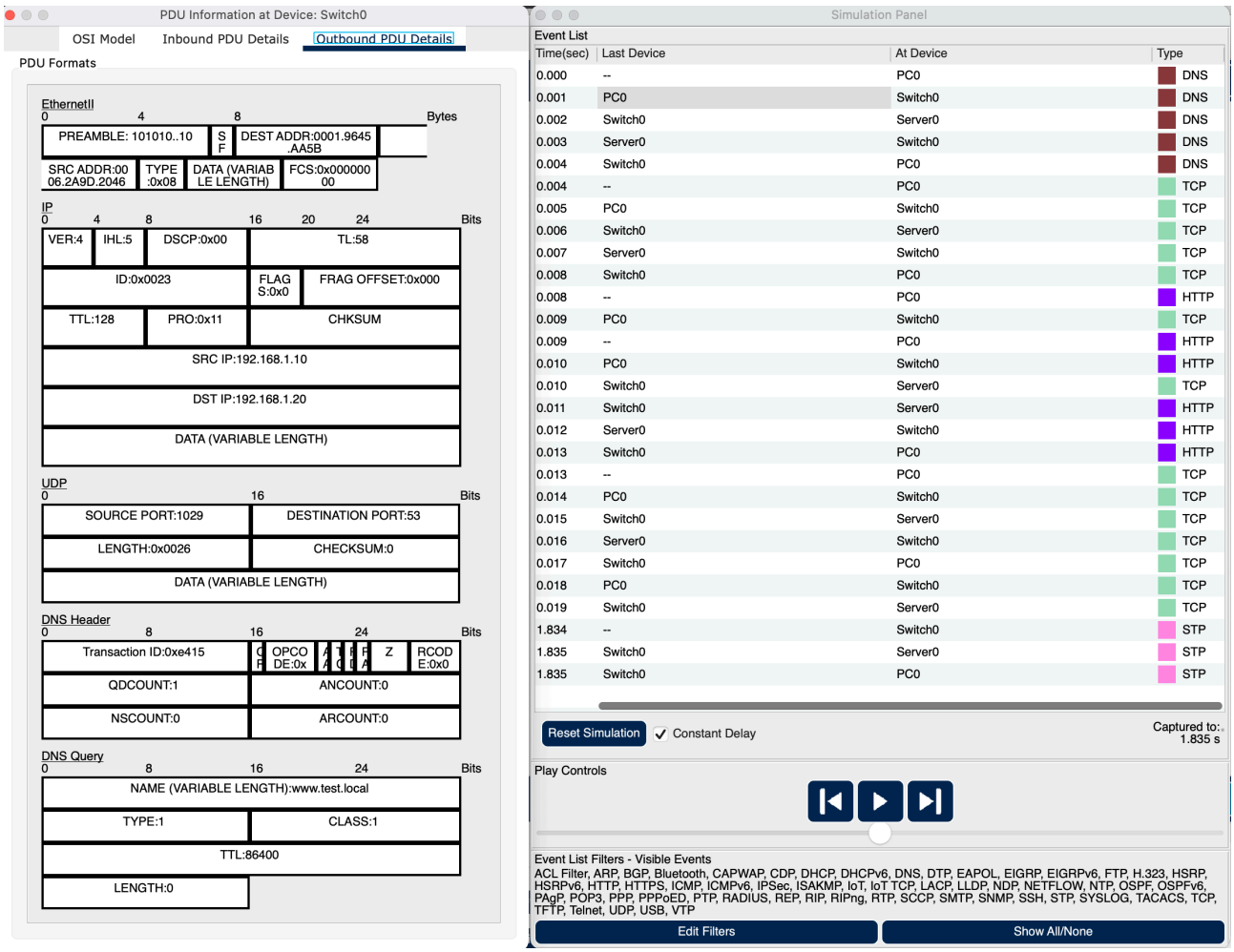
Once the domain name is resolved, the client establishes a TCP connection with the web server.

First, the three-way handshake process (SYN, SYN-ACK, ACK) occurs to establish the connection.
Then, the client sends an HTTP request to retrieve the web page hosted on the server.

The Hypertext Transfer Protocol uses:

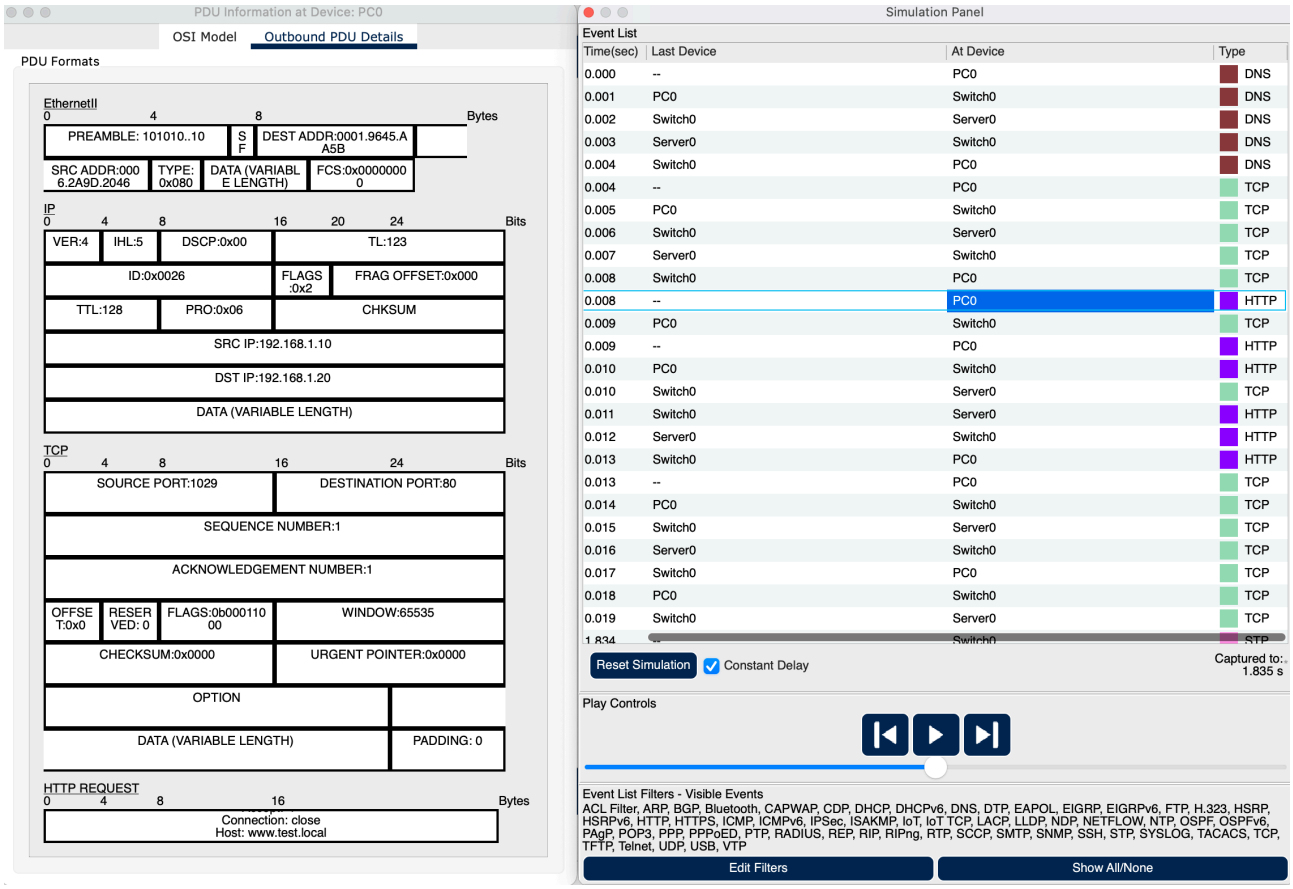
TCP port 80

Figure 5 – DNS Query



DNS query from the client to the server to resolve the domain name www.test.local. The exchange of request and response packets is observed using the DNS protocol over port 53.

Figure 6 – HTTP Request



4.4 FTP Session

The client establishes a File Transfer Protocol session with the server to authenticate and access the file transfer service.
During the FTP session, the client authenticates using a username and password and then sends commands to interact with the server.

The authentication process is performed using the following commands:

USER – sends the username
PASS – sends the password
LIST – retrieves the list of files

These commands allow authentication and access to the files available on the server.

The File Transfer Protocol uses:

TCP port 21

Figure 7 – FTP Session

FTP session from the PC to the server showing successful authentication and file listing using the DIR command.

```
Connected to 192.168.1.20
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.1.20:
0   : asa842-k8.bin                                    5571584
1   : asa923-k8.bin                                    30468096
2   : c1841-advipservicesk9-mz.124-15.T1.bin           33591768
3   : c1841-ipbase-mz.123-14.T7.bin                    13832032
4   : c1841-ipbasek9-mz.124-12.bin                     16599160
5   : c1900-universalk9-mz.SPA.155-3.M4a.bin           33591768
6   : c2600-advipservicesk9-mz.124-15.T1.bin           33591768
7   : c2600-i-mz.122-28.bin                            5571584
8   : c2600-ipbasek9-mz.124-8.bin                      13169700
9   : c2800nm-advipservicesk9-mz.124-15.T1.bin         50938004
10  : c2800nm-advipservicesk9-mz.151-4.M4.bin          33591768
11  : c2800nm-ipbase-mz.123-14.T7.bin                  5571584
12  : c2800nm-ipbasek9-mz.124-8.bin                    15522644
13  : c2900-universalk9-mz.SPA.155-3.M4a.bin           33591768
14  : c2950-i6q412-mz.121-22.EA4.bin                   3058048
15  : c2950-i6q412-mz.121-22.EA8.bin                   3117390
16  : c2960-lanbase-mz.122-25.FX.bin                   4414921
17  : c2960-lanbase-mz.122-25.SEE1.bin                 4670455
18  : c2960-lanbasek9-mz.150-2.SE4.bin                 4670455
19  : c3560-advipservicesk9-mz.122-37.SE1.bin          8662192
20  : c3560-advipservicesk9-mz.122-46.SE.bin           10713279
21  : c800-universalk9-mz.SPA.152-4.M4.bin             33591768
22  : c800-universalk9-mz.SPA.154-3.M6a.bin            83029236
23  : cat3k_caa-universalk9.16.03.02.SPA.bin           505532849
24  : cgr1000-universalk9-mz.SPA.154-2.CG              159487552
25  : cgr1000-universalk9-mz.SPA.156-3.CG              184530138
26  : ir800-universalk9-bundle.SPA.156-3.M.bin         160968869
27  : ir800-universalk9-mz.SPA.155-3.M                 61750062
28  : ir800-universalk9-mz.SPA.156-3.M                 63753767
29  : ir800_yocto-1.7.2.tar                            2877440
30  : ir800_yocto-1.7.2_python-2.7.3.tar               6912000
31  : pt1000-i-mz.122-28.bin                           5571584
32  : pt3000-i6q412-mz.121-22.EA4.bin                  3117390
ftp>
```

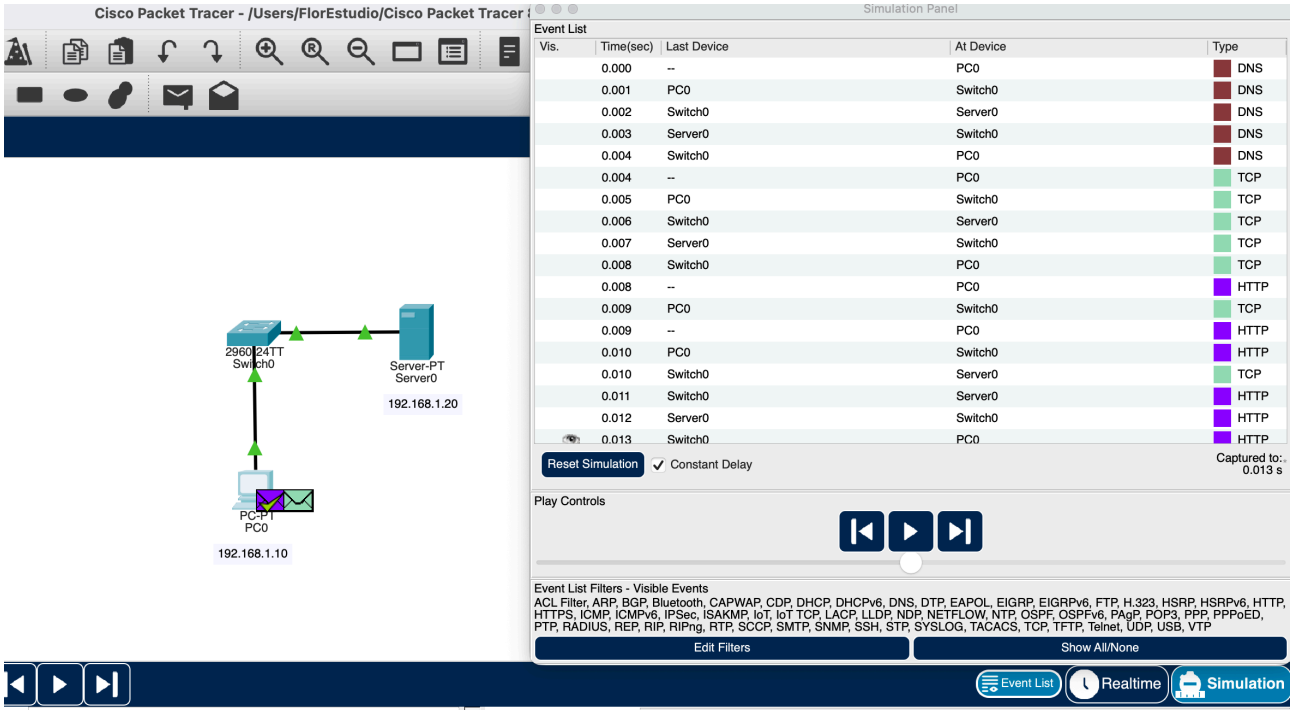
4.5 Event List Analysis

During the simulation, it was possible to observe the different protocols involved in the communication between the client and the server in the Event List.
This list shows the sequence of events generated during access to DNS, HTTP, and FTP services, allowing the identification of the protocols and ports used by each service.

The events are displayed in chronological order (Time sec), enabling analysis of the communication sequence between devices.

The Event List shows the sequence of events generated during the communication between the client and the server. Packets from Domain Name System are observed for domain resolution, followed by connection establishment using Transmission Control Protocol, and finally web page transfer using Hypertext Transfer Protocol.

Figure 8 – Event List in Simulation Mode



The Event List displays the sequence of events generated during communication between the client and the server. Domain Name System packets are observed for domain name resolution, followed by connection establishment using Transmission Control Protocol, and finally the web page transfer using Hypertext Transfer Protocol.

5. Results

During the laboratory, the following results were obtained:

- Successful name resolution using DNS

- HTTP connection successfully established

- Successful authentication in the FTP service

- File listing retrieved from the FTP server

- Identification of ports used by each service

- Observation of packet exchange using Simulation Mode

6. Conclusion

This laboratory allowed the analysis of how different application layer protocols interact within a local network.

The DNS resolution process, TCP connection establishment for HTTP, and authentication in an FTP session were observed.

The use of Simulation Mode in Cisco Packet Tracer made it possible to visualize the exchanged packets and understand the use of ports and transport protocols.

A clear understanding was achieved regarding the relationship between application protocols, transport protocols, and port usage in client-server communication.